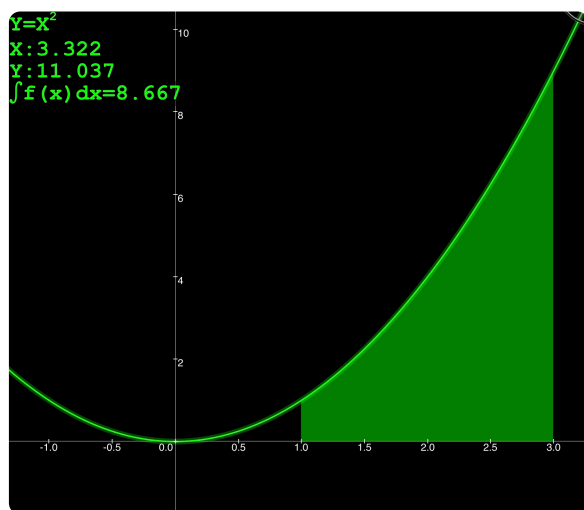


Section 5.4 Integrals and Net Change

Definite Integral: $\int_1^3 x^2 dx$ is used to

find the area under $f(x) = x^2$ on $[1, 3]$.

$$\int_1^3 x^2 dx = \left. \frac{x^3}{3} \right|_1^3 = \frac{3^3}{3} - \frac{1}{3} = \boxed{\frac{26}{3}}$$

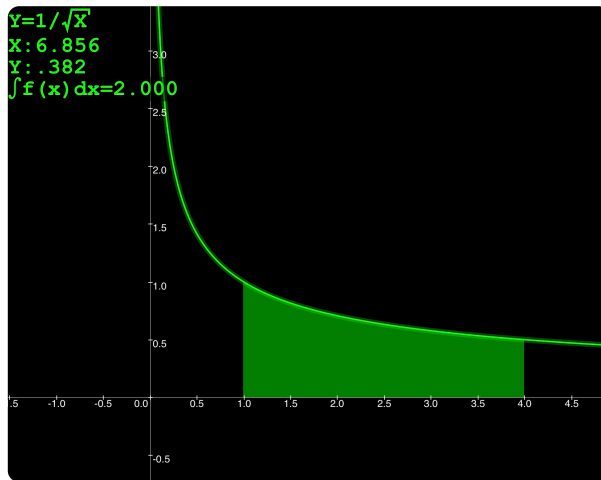


Indefinite Integral $\int x^2 dx = \frac{x^3}{3} + c$

↑
The antiderivative
of x^2

$$\underline{\text{Ex}} \quad ① \quad \int_1^4 \frac{1}{\sqrt{x}} dx = \int_1^4 x^{-\frac{1}{2}} dx$$

$$= 2x^{\frac{1}{2}} \Big|_1^4 = 2\sqrt{4} - 2\sqrt{1} = \textcircled{2}$$



$$\underline{\text{Ex}} \quad ② \quad \int_0^1 \frac{1}{1+x} dx$$

$$= \ln(1+x) \Big|_0^1$$

$$= \ln(2) - \ln(1)$$

$$= \textcircled{0.693}$$

Ex ③ $\int \frac{1}{(1+x)^2} dx$

$$= \int (1+x)^{-2} dx$$

$$= \frac{(1+x)^{-1}}{-1} + C$$

$$= -\frac{1}{1+x} + C$$

Ex ④ $\int \left(\frac{x^3 + \sqrt{x} + 7}{x} \right) dx$

$$= \int \left(\frac{x^3}{x} + \frac{\sqrt{x}}{x} + \frac{7}{x} \right) dx$$

$$= \int \left(x^2 + x^{-\frac{1}{2}} + 7 \cdot \frac{1}{x} \right) dx$$

$$= \frac{x^3}{3} + 2x^{\frac{1}{2}} + 7 \cdot \ln x + C$$

$$\underline{\text{Ex}} \quad (5) \quad \int_0^{\frac{\pi}{4}} \frac{1 + \cos^2 x}{\cos^2 x} dx$$

$$= \int_0^{\frac{\pi}{4}} \left(\frac{1}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} (\sec^2 x + 1) dx$$

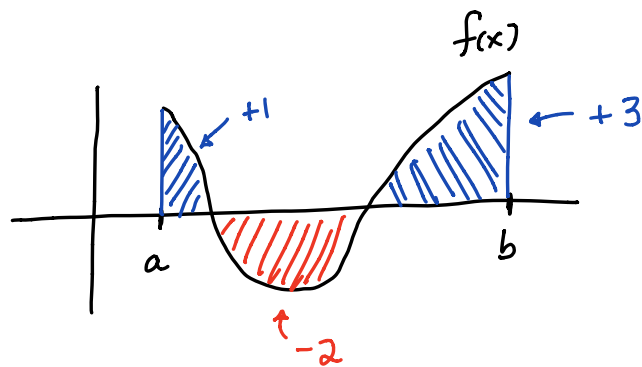
$$= \tan x + x \Big|_0^{\frac{\pi}{4}}$$

$$= \tan\left(\frac{\pi}{4}\right) + \frac{\pi}{4} - (\tan 0 - 0)$$

$$= 1 + \frac{\pi}{4}$$

Net Change $\int_a^b f(x) dx$

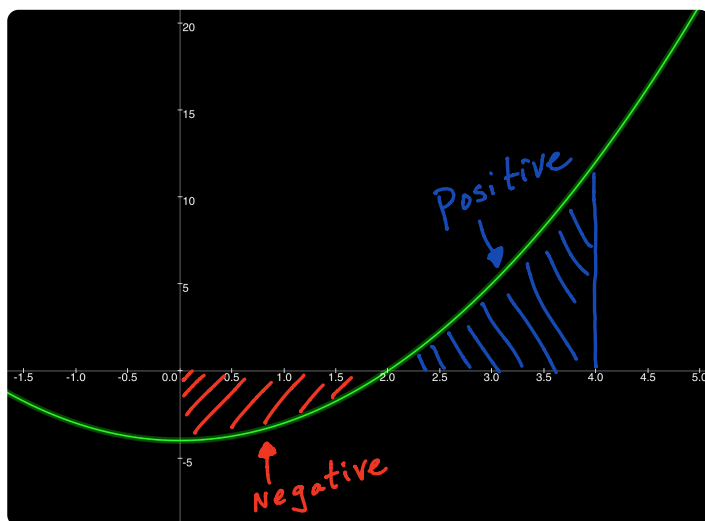
Suppose that $f(x)$ is both positive and negative on $[a, b]$.



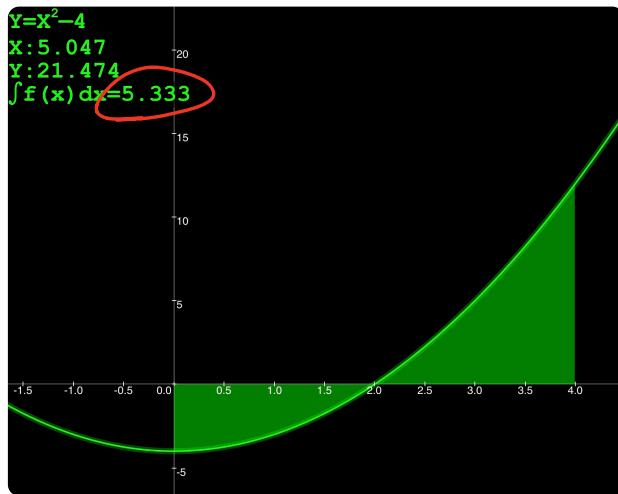
$$\text{Net Change} = +1 - 2 + 3 = \boxed{+2}$$

$$\text{So, } \int_a^b f(x) dx = \boxed{2}$$

Ex ⑥ Let $f(x) = x^2 - 4$ on $[0, 4]$



$$\begin{aligned}
 \text{Net Change} &= \int_0^4 (x^2 - 4) dx \\
 &= \left. \frac{x^3}{3} - 4x \right|_0^4 \\
 &= \frac{4^3}{3} - 4(4) - \left(\frac{0^3}{3} - 4(0) \right) \\
 &= \frac{64}{3} - 16 = \underline{5.333}
 \end{aligned}$$



Ex ⑦ Drop a rock from the top of a building. The velocity will be $v(t) = -32t$, in feet/sec. Evaluate $\int_0^1 v(t) dt$ and explain its meaning in this situation.

$$\begin{aligned}\int_0^1 -32t \cdot dt &= -32 \frac{t^2}{2} \Big|_0^1 \\&= -16t^2 \Big|_0^1 \\&= -16(1) - (-16(0)) \\&= -16\end{aligned}$$

From $t = 0$ seconds to $t = 1$ second,
the rock dropped 16 feet.